

Aerodynamic Shape Optimization for the NURBS-Enhanced Discontinuous Galerkin Method

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I. Introduction

Design in aerospace settings has undergone significant changes in the past decades, motivated primarily by massive improvements in computing resources as well as increased research into numerical schemes and mathematical models. In the field of Computational Fluid Dynamics (CFD), we have seen the development of faster and more efficient algorithms for computationally resolving flows over complicated geometries. These improvements in CFD, coupled with the development of numerical methods and algorithms for solving complicated constrained optimization problems, have greatly improved the ability to generate optimal designs in practical aerospace applications [1].

In the field of CFD, one area that has received increased research in past years has been in numerical schemes known as high-order methods. Equipped with higher spatial orders of accuracy, these methods are able to resolve numerical solutions with much higher accuracy, relative to traditional low-order methods, using the same number of degrees of freedom. Past years have seen several high-order methods be the principal focus of research, including the Discontinuous Galerkin method [2, 3], Spectral Difference method [4, 5] and Flux Reconstruction approach [6–8]. While these schemes have been studied in great detail, their use in aerodynamic optimization is at a preliminary stage [9–11].

Design in aerospace settings has also encountered significant changes. Prior to the spread of efficient numerical methods and optimizers, aerodynamic design had largely been an analytical process as optimal shapes were arrived at by using mathematical solutions to simplified problems [12]. As computational resources improved, numerical schemes were developed to solve these optimization problems. One such class of schemes were gradient-based optimization methods which require the gradient of a specified cost function, which must be minimized, for the optimization. The drawback, however, is that using traditional methods this gradient is often quite costly to compute. Fortunately, the adjoint approach provides a remedy for this. Developed using control theory, the adjoint method allows for an efficient computation of the gradient by calculating it indirectly through solving what is termed as the adjoint equation. The process results in an efficient computation of the gradients, with a computational cost that is independent of the number of design variables and is as expensive as one flow solve [12]. The adjoint approach has become ubiquitous in gradient-based design, as it has not only successfully been used in numerous aerospace applications such as in transonic design problems, but it has also been developed extensively for commercial CFD applications.

Although CFD and aerodynamic optimal design has experienced substantial changes and improvements, one bottleneck that still remains in the design process today is in interfacing with Computer Aided Design (CAD). Designers, presently, first work exclusively with CAD to generate geometric models for a given engineering application. In engineering design, these models are primarily represented in CAD programs using Non-Uniform Rational B-Splines (NURBS) [13], a popular choice as the NURBS basis allows for convenient free-form surface modeling, is able to represent conic sections exactly and is mature enough that there exist many efficient and stable algorithms for NURBS object generation [14]. Upon completing a first design in a CAD software, designers may then move to validate their model by performing numerical simulations using an analysis software (to compute a CFD or perhaps FEA simulation). Here, complications quickly arise, as the geometric model must first be approximated and translated into an analysis-suitable geometry for the analysis software and, second, significant time and effort must be devoted to generating the mesh [14]. The approximation of the CAD geometry, moreover, results in an inevitable loss of accuracy, as the analysis is completed on an inexact representation of the model. Designers may also go one step further and perform a shape optimization in the given analysis software. Here, once again, a pitfall is encountered as the final optimal design in the analysis software must be approximated once again, this time using NURBS basis functions, in order to transfer the optimized geometry back into a CAD representation.

The complications resulting from CAD interactions have resulted in increased research into approaches to alleviate

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these problems. A novel approach, presented by Hughes, involves the use of isogeometric analysis (IGA), where the principal idea is to use the same set of basis functions used for the CAD representation of the geometry, the NURBS basis, for the analysis as well [14]. The use of NURBS as a basis for analysis allows, in turn, for an exact representation of the geometry during the simulation. Some limitations do still exist for IGA however, including difficulties in meshing resulting from the need for structured, tensor-product meshes as well as the significant effort required to implement IGA into existing finite element codes [15].

As an alternative approach, we consider in this work the NURBS-enhanced Finite Element method (NEFEM). Analogous to IGA, NEFEM can exactly parametrize CAD geometries represented using NURBS basis functions. This is accomplished by using the NURBS description of the boundary to compute all geometry related metric terms for the analysis. NEFEM, moreover, differs from IGA in that it places no restriction on the function space used for the approximation of the solution. As such, standard polynomial basis functions, as opposed to the NURBS basis as incorporated in IGA, may be used to resolve the numerical solution [16].

In this work, we demonstrate aerodynamic shape optimization results using the NURBS-enhanced approach and investigate optimization problems using the high-order Discontinuous Galerkin (DG) method. Furthermore, the adjoint approach has been used to efficiently compute all gradients needed for the optimization, with the optimization completed using a Sequential Quadratic Programming method. Finally, we demonstrate optimal solutions to both unconstrained and constrained shape optimization problems on NURBS geometries.

II. The NURBS-Enhanced Discontinuous Galerkin Method

In this section, first an overview will be given of the Discontinuous Galerkin (DG) method, with a particular focus on solving the scalar conservation equation. For a more detailed formulation of this method, we refer readers to the textbook by Hesthaven and Warburton [17]. Following the implementation details for the DG method, the NURBS-enhanced approach is presented.

A. The Discontinuous Galerkin Method

Consider the general 2D scalar conservation equation given by

$$\begin{aligned} \frac{\partial}{\partial t} U(\mathbf{x}, t) + \nabla \cdot \mathbf{F}(U(\mathbf{x}, t)) &= 0 \\ U(\mathbf{x}, 0) &= U_0(\mathbf{x}) \\ t \geq 0, \quad \mathbf{x} \in \Omega \end{aligned} \tag{1}$$

where $\mathbf{x} = [x \ y]$ corresponds to the Cartesian coordinates, Ω corresponds to the physical domain and $\mathbf{F}(U(\mathbf{x}, t)) = [F_1(U(\mathbf{x}, t)) \ F_2(U(\mathbf{x}, t))]$ consists of the fluxes in each Cartesian coordinate direction. The numerical solution to equation (1) is sought over a computational domain Ω^h , $\Omega^h \approx \Omega$, partitioned into N_e non-overlapping elements such that

$$\Omega^h = \bigcup_{n=1}^{N_e} \Omega_n. \tag{2}$$

In equation (2), Ω_n corresponds to the elements on the domain, each of which has an outer surface denoted by Γ_n . With the domain discretized, the solution to equation (1) is considered within each element. Let the approximate numerical solution to the PDE in equation (1) within the k^{th} element, Ω_k , be $U_k^h(\mathbf{x}, t)$, where $U_k^h(\mathbf{x}, t)$ is defined to be zero outside of the element. Using the local solution from each element, the global approximate solution, $U^h(\mathbf{x}, t)$, over the computational domain, Ω^h , can be written as

$$U^h(\mathbf{x}, t) = \bigoplus_{k=1}^{N_e} U_k^h(\mathbf{x}, t). \tag{3}$$

In the DG approach, rather than directly searching for the numerical solution over the whole domain, the solution on each individual element will be computed separately. Furthermore, it is important to note that no restriction on continuity will be placed on the solutions between these elements and, as such, multiple solutions may be defined on element boundaries.

The solution in each element is first represented using a set of basis functions. Consider the k^{th} element Ω_k . On this element, define N_p linearly independent modal or nodal basis functions $\psi_k(\mathbf{x})$ of maximum order P which are identically zero outside of Ω_k . The numerical solution U_k^h on this element can then be represented as a linear combination of these basis functions and can be written as

$$U_k^h(\mathbf{x}, t) = \sum_{i=1}^{N_p} U_{k,i}(t) \psi_i(\mathbf{x}). \quad (4)$$

The number of basis functions used to resolve the numerical solution varies based on the element shape as well as the order P . For instance, for tensor product elements (such as quadrilaterals and hexahedral elements), $N_p = (P + 1)^d$, where d corresponds to the dimension of the physical domain.

The residual can now be expressed on each element using the approximate numerical solution and is given, on the k^{th} element, as

$$R_k^h(\mathbf{x}, t) = \frac{\partial}{\partial t} U_k^h(\mathbf{x}, t) + \nabla \cdot \mathbf{F}(U_k^h(\mathbf{x}, t)). \quad (5)$$

This residual is required to be orthogonal to each basis function, thus yielding N_p equations of the form:

$$\int_{\Omega_k} R_k^h(\mathbf{x}, t) \cdot \psi_i(\mathbf{x}) d\Omega = 0, \quad \forall i = 1, \dots, N_p. \quad (6)$$

Substituting the expression for the residual from equation (5) into equation (6) and integrating by parts, the weak form of the DG scheme is obtained:

$$\begin{aligned} \int_{\Omega_k} \psi_i(\mathbf{x}) \cdot \frac{\partial}{\partial t} U_k^h(\mathbf{x}, t) d\Omega_k - \int_{\Omega_k} \nabla \psi_i(\mathbf{x}) \cdot \mathbf{F}(U_k^h(\mathbf{x}, t)) d\Omega_k + \\ \int_{\Gamma_k} \psi_i(\mathbf{x}) \cdot (\mathbf{F}(U_k^h(\mathbf{x}, t)) \cdot \hat{\mathbf{n}})^* d\Gamma_k = 0, \quad \forall i = 1, \dots, N_p. \end{aligned} \quad (7)$$

Equation (7) consists of two volume integral terms over the element Ω_k and a surface integral term over the boundary Γ_k of the element. Furthermore, the expression $(\mathbf{F}(U_k^h(\mathbf{x}, t)) \cdot \hat{\mathbf{n}})^*$ corresponds to the numerical flux.

To efficiently obtain the numerical solution to the PDE on the physical domain Ω , a transformation mapping is used. This transformation maps each given element on the physical domain onto a unique reference element Ω_r , with a boundary Γ_r , on a domain specified using a set of reference coordinates $\hat{\xi}$ (Figure (1)). The choice of the reference element can vary, and is dependent inherently on the type of the physical element in the mesh. For instance, if a problem is considered on a two-dimensional physical domain consisting of a mesh with quadrilateral elements, the quadrilateral reference element is given by

$$\Omega_r := \{\hat{\xi} := (\hat{\xi}, \hat{\eta}) \mid -1 \leq \hat{\xi}, \hat{\eta} \leq 1\}. \quad (8)$$

The transformation mapping from the reference domain to the physical domain can moreover, in general, be expressed as

$$\mathbf{x}(\hat{\xi}) = \Theta(\hat{\xi}). \quad (9)$$

The mapping in equation (9) can be substituted into the weak form of the DG scheme in equation (7) and all integrals can be mapped from the physical domain onto the reference element. The mapped version of the equation is then given by

$$\begin{aligned} \int_{\Omega_r} \psi_i(\Theta(\hat{\xi})) \cdot \frac{\partial}{\partial t} U_k^h(\Theta(\hat{\xi}), t) \cdot J_k^\Omega d\Omega_r - \int_{\Omega_r} \hat{\nabla} \psi_i(\Theta(\hat{\xi})) \cdot \hat{\mathbf{F}}(U_k^h(\Theta(\hat{\xi}), t)) \cdot J_k^\Omega d\Omega_r + \\ \int_{\Gamma_r} \psi_i(\Theta(\hat{\xi})) \cdot (\mathbf{F}(U_k^h(\Theta(\hat{\xi}), t)) \cdot \hat{\mathbf{n}})^* \cdot J_k^\Gamma d\Gamma_r = 0, \quad \forall i = 1, \dots, N_p, \end{aligned} \quad (10)$$

where J_k^Ω and J_k^Γ correspond to the Jacobian of the volume and surface integrals mapped onto the reference element, respectively, $\hat{\nabla}$ represents the gradient operator on the reference domain and $\hat{\mathbf{F}}$ denotes the flux mapped onto the

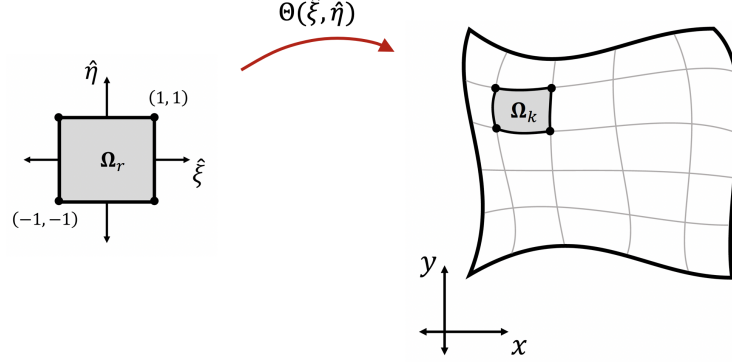


Fig. 1 An example of the mapping of the standard reference element on the reference domain to an arbitrary element on the physical domain.

reference domain. As a final step, equation (4) may be substituted into the first volume integral, and the summation and integration may be interchanged to yield

$$\sum_{j=1}^{N_p} \frac{d}{dt} U_{k,j}(t) \left(\int_{\Omega_r} \psi_i(\Theta(\hat{\xi})) \cdot \psi_j(\Theta(\hat{\xi})) \cdot J_k^\Omega d\Omega_r \right) - \int_{\Omega_r} \hat{\nabla} \psi_i(\Theta(\hat{\xi})) \cdot \hat{F}(U_k^h(\Theta(\hat{\xi}), t)) \cdot J_k^\Omega d\Omega_r + \int_{\Gamma_r} \psi_i(\Theta(\hat{\xi})) \cdot (F(U^h(\Theta(\hat{\xi}), t)) \cdot \hat{n})^* \cdot J_k^\Gamma d\Gamma_r = 0, \quad \forall i = 1, \dots, N_p. \quad (11)$$

The integrals in equation (11) may be evaluated using Gaussian quadrature, thus yielding a system of N_p equations for the N_p unknown coefficient derivatives ($\frac{dU_{k,j}}{dt}$) on the k^{th} element. This linear system can be set up for each individual element and solved at each time step for the derivative terms. If an explicit scheme is used, the values of the derivatives can be used with a time stepping scheme (such as a Runge Kutta or Forward Euler scheme) to obtain the values of the coefficients $U_{k,j}$ for the subsequent time step in order to traverse the numerical solution in time.

B. Metric Computations

Central to the process of transforming the integrals, gradients and fluxes from the physical domain to the reference domain in equation (10) are the metric terms which are computed using the transformation in equation (9). Consider a physical domain of dimension d so that $\mathbf{x} \in \mathbb{R}^d$ and $\hat{\xi} \in \mathbb{R}^d$. The metric terms are then given as

$$\frac{\partial x_i}{\partial \hat{\xi}_j}, \quad \forall i, j = 1, \dots, d. \quad (12)$$

With the metric terms, it is then straightforward to compute the Jacobian terms in the mapped volume and surface integrals, the normal vector terms for element faces on the physical domain, as well as map the gradient operator and flux to the reference domain. In addition, metric identities must be evaluated in a manner to satisfy freestream-preservation. In [18], Kopriva provides the necessary conditions to ensure that the scheme satisfies this condition. In summary, the common cross-product form only satisfy the condition in two-dimensions but not three, where the author provides a curl form of the metric terms which always satisfy and preserve freestream flow. For further details regarding this mapping process, we refer the reader to [19].

The transformation to map each element on the physical domain to the reference element on the reference domain was briefly introduced in the previous section through equation (9). In this section, a more detailed explanation of this transformation function, and how the function may be used to compute the metric terms, will be given. First, what will be denoted as the standard approach to element mapping will be outlined. Following this, an alternative method known as the NURBS-enhanced approach, which is the principle focus of this work, is presented.

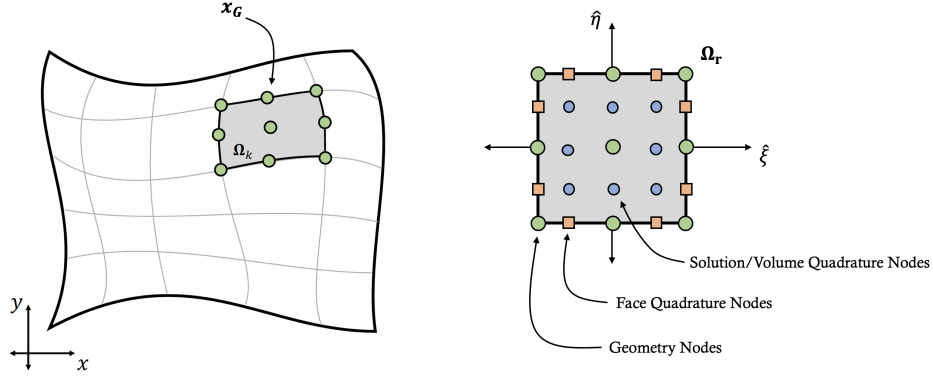


Fig. 2 The mapping of a standard quadrilateral reference element, Ω_r , on the reference domain to an arbitrary quadrilateral element on the physical domain, Ω_k using the standard or traditional approach.

1. The Standard Approach

In the traditional or standard approach, the mapping of each element on the physical domain onto the reference domain is completed using a set of shape functions and geometry control points. The mapping function may then be written as

$$\mathbf{x}(\hat{\xi}) = \Theta(\hat{\xi}) = \sum_{i=1}^{N_G} \mathbf{x}_G \cdot \phi_i(\hat{\xi}), \quad (13)$$

where $\mathbf{x}_G$ correspond to the geometry control points and $\phi_i(\hat{\xi})$ denotes the shape functions. Moreover, the variable N_G refers to the number of geometry control points and shape functions used for the mapping, which may be increased to generate a more accurate parametrization of the geometry.

Several different schemes may be employed to map the physical elements to and from the reference domain. The class of sub-parametric, isoparametric and super-parametric representations is the most popular and involves using as the shape functions for the mapping the same space of basis functions used to represent the numerical solution (given in equation (4)). The distinction between each parametric representation is given by the number of shape functions and geometry control points, N_G , used for the mapping; for isoparametric representations the number of shape functions is equal to the number of basis functions used to represent the solution in each element ($N_G = N_P$) whereas in the sub-parametric case $N_G < N_P$ and in the super-parametric case $N_G > N_P$. Furthermore, with the element mapping defined in equation (13), the metric terms can be readily computed using the partial derivatives of the set of shape functions $\phi_i(\hat{\xi})$ (defined using a set of geometry nodes on the reference element) with respect to the reference domain spatial coordinates $\hat{\xi}$. These metric terms can then be interpolated onto the face and volume quadrature nodes in order to numerically compute all integrals. This process is summarized in the schematic shown in Figure (2).

The use of isoparametric, sub-parametric and super-parametric transformations is prevalent in various high-order codes. The major shortcoming of this approach, however, is in the accuracy of the representation of the geometry. Classically, high-order methods employ polynomial modal or nodal basis functions for resolving the numerical solution. This, in turn, results in this standard transformation approach to utilize shape functions that also belong to the space of polynomial functions. For practical engineering design applications, the geometrical model is typically represented in CAD using NURBS basis functions and, as a result of this, a geometry parametrization using polynomial shape functions will result in a loss of accuracy due to an inexact representation of the geometry.

2. The NURBS-Enhanced Approach

To illustrate the NURBS-enhanced approach, a brief overview of NURBS geometries is needed. NURBS geometries consist of a mapping $\tilde{\Theta}$ from a NURBS parametric domain to the physical domain of the form $\tilde{\Theta} : \mathbb{R}^d \rightarrow \mathbb{R}^d$. The mapping consists of the NURBS basis functions (a multivariate NURBS basis is used if the dimension is greater than 1) and the control points. For instance, a 2D NURBS geometric model (known as a NURBS patch) is defined through a mapping of the form

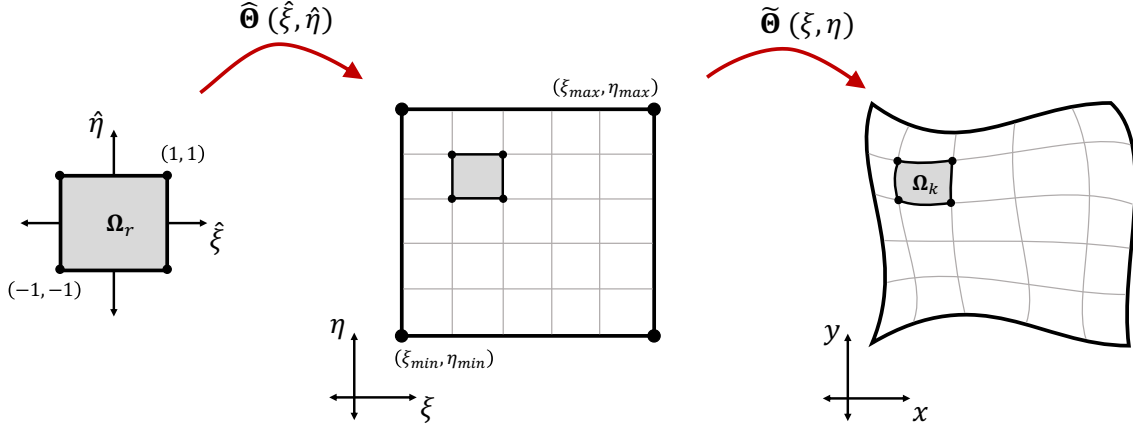


Fig. 3 The mapping of a standard quadrilateral reference element, Ω_r , on the reference domain to an arbitrary quadrilateral element on the physical domain, Ω_k using the NURBS-enhanced approach. Shown in the middle is the NURBS parametric domain and on the right is the physical domain with the NURBS geometric model which, in this two-dimensional case, is a NURBS patch.

$$\tilde{\Theta}(\xi_1, \xi_2) = \sum_i \sum_j R_{i,j}(\xi_1, \xi_2) \cdot \mathbf{P}_{i,j}, \quad (14)$$

where ξ_1 and ξ_2 are the NURBS parametric domain coordinates, $\mathbf{P}_{i,j}$ refer to the control points and $R_{i,j}(\xi_1, \xi_2)$ are the NURBS basis functions. For further details regarding the NURBS basis functions and NURBS geometries, we refer the reader to [13].

The NURBS-enhanced approach aims to address the central shortcoming of the standard approach in regards to its error in representing CAD geometric models parametrized using NURBS basis functions. In this approach, the geometric model, and its NURBS parametrization, is used directly to compute all metric terms needed for mapping elements on the physical domain to and from the reference domain. This ultimately allows for an exact parametrization of the geometry in a finite element framework. Moreover, although NURBS basis functions may be used for computing all geometry related parameters, the NURBS-enhanced approach does not restrict the space of functions to use for resolving the numerical solution. Thus, standard modal or nodal polynomial basis functions may still be used to represent the solution.

The transformation of elements from the reference domain onto the physical domain is a two step process for the NURBS-enhanced approach involving two transformation functions of the form:

$$\begin{aligned} \xi(\hat{\xi}) &= \hat{\Theta}(\hat{\xi}), \\ \mathbf{x}(\xi) &= \tilde{\Theta}(\xi). \end{aligned} \quad (15)$$

In the first step, the reference element, Ω_r , on the reference domain with spatial coordinates $\hat{\xi}$ is mapped to the NURBS parametric domain, spanned by the spatial coordinates ξ , using a linear mapping $\hat{\Theta}$. In the second step, the element on the NURBS parametric domain is mapped to the physical space using a NURBS mapping, $\tilde{\Theta}$, of the form given in equation (14). This mapping consists of the NURBS basis functions and the set of NURBS control points defining the NURBS geometric model in the physical space. Figure (3) contains a schematic depicting the steps of this mapping process for an arbitrary two-dimensional case.

The computation of the metric terms can furthermore be completed using both transformation functions and the chain rule. Consider a physical domain of dimension d . The metric terms can then be given by

$$\frac{\partial x_i}{\partial \hat{\xi}_j} = \sum_{k=1}^d \frac{\partial x_i}{\partial \xi_k} \cdot \frac{\partial \xi_k}{\partial \hat{\xi}_j}, \quad \forall i, j = 1, \dots, d. \quad (16)$$

The terms of the form $\frac{\partial x_i}{\partial \xi_k}$ can be computed exactly using the derivatives of the NURBS geometry parametrization, and is found using the NURBS control points and the derivatives of the NURBS basis functions with respect to the NURBS parametric domain spatial coordinates ξ . Moreover, the terms of the form $\frac{\partial \xi_k}{\partial \xi_j}$ can be easily computed using the linear mapping of the element from the reference domain onto the NURBS parametric domain. We once again emphasize that, with the metric terms computed using the NURBS-enhanced approach, the NURBS geometric model is able to be parametrized exactly when computing the numerical solution.

III. Optimization Framework

The optimization algorithm, based on the adjoint approach, is detailed in this section. Consider the case where a cost function I is to be minimized using a set of N_d design variables α where $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_{N_d}]$. In gradient-based optimization approaches, in order to minimize the cost function, its gradient with respect to these N_d design variables is needed:

$$\nabla_{\alpha} I = \left[\frac{\partial I}{\partial \alpha_1}, \dots, \frac{\partial I}{\partial \alpha_{N_d}} \right]. \quad (17)$$

The adjoint approach allows for the ability to efficiently compute the gradient of the given objective function with a computational cost as expensive as one flow solve. The strength, therefore, of the approach is that its cost is independent of the number of design variables used for the optimization.

Consider the objective function $I(\mathbf{w}, \alpha)$, where I is a function of the N_f flow variables $\mathbf{w}$ and the N_d design variables α . These flow variables must additionally satisfy the flow constraint equations

$$\mathbf{R} = \mathbf{R}(\mathbf{w}, \alpha) = 0. \quad (18)$$

The flow constraint equations, $\mathbf{R}$, is a vector-valued function with N_f components that consists of the physical equations in their discretized form. The adjoint method to compute the gradient involves a two step process. In the first step, the adjoint equation is solved for the adjoint, ψ :

$$\left[\frac{\partial \mathbf{R}}{\partial \mathbf{w}} \right]^T \psi = \left[\frac{\partial I}{\partial \mathbf{w}} \right]. \quad (19)$$

In the second step, the adjoint ψ is substituted into the sensitivity equation to compute the gradient of the functional:

$$\nabla_{\alpha} I = \left[\frac{\partial I}{\partial \alpha} \right]^T - \psi^T \left[\frac{\partial \mathbf{R}}{\partial \alpha} \right]. \quad (20)$$

The computed gradient may finally be input into an optimizer to then obtain the value of the design variables for the next design iteration. This process may then be repeated until an optimal solution is attained after several design iterations. For further details, we refer readers to [20].

Once the geometry is deformed after a given design cycle, the mesh must be adjusted. Many approaches exist to tackle this problem. Mesh regeneration, after each design cycle, is one solution, however it quickly becomes impractical due to the increased computational complexity involved. Due to this, mesh movement schemes have received increased attention in the past few years, where surface perturbations are propagated through the domain to deform the grid. Methods for mesh deformation include linear cell analogy methods [21] and radial basis function deformation schemes and their extensions [22] to name a few.

The NURBS-enhanced approach addresses the problem of mesh deformation automatically through the use of NURBS patches to parametrize the domain. Recall each element in the mesh is mapped from the parametric domain to the physical domain through the NURBS mapping. Furthermore, any change in the surface induced by the displacement of the control points will result in a change in the mapping. This, in turn, will automatically reposition the elements on the physical domain, resulting in no need for an additional mesh deformation algorithm to propagate the surface deformation into the volume mesh.

IV. Numerical Validation

To validate the implementation of the NURBS-enhanced DG solver, this section presents a mesh refinement study for a canonical case involving flow through a channel with a bump.

The case that is considered to validate the orders of accuracy is for flow through an inviscid 2D channel parametrized using a NURBS patch (shown in Figure (4)). This channel consists of a straight upper face, while on the lower face a bump is introduced. A slip wall boundary condition is placed first on the upper and lower faces of the channel. A total temperature, T_T , and total pressure, p_T , are further specified on the leftmost boundary of the mesh. Moreover, a fixed back pressure is specified on the right boundary of the mesh with a value of $p_{back} = 0.99 p_T$, resulting in subsonic flow from left to right. Figure (4) depicts the entropy error convergence results. It can be seen that the optimal, and expected, convergence order of $P + 1$ is obtained.

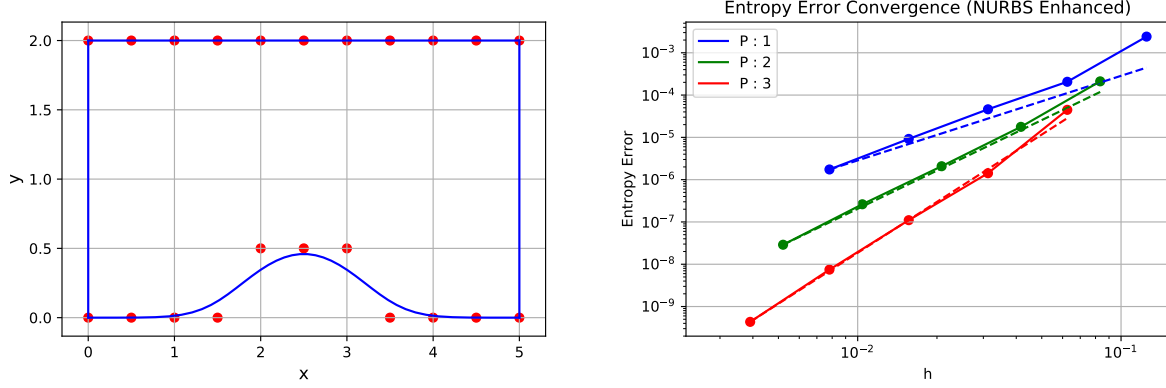


Fig. 4 The NURBS patch for the channel with the bump (left). Shown in blue is the outline for the patch defined by the NURBS control points (shown in red). The entropy error convergence for $P = 1, 2, 3$ for the subsonic flow through the channel (right). Reference lines with slope $P + 1$ are shown in the dashed lines.

V. Optimization Results

This section presents two optimization cases incorporating the NURBS-enhanced DG method. An inverse design case is first introduced followed by a target lift coefficient optimization case.

A. NACA 4412 Inverse Design

The inverse design problem consists of having a given geometry deform from its initial shape to a desired target shape. This is done through specifying a target pressure distribution to be attained, $p_{target}(s)$, which is given by the pressure over the target shape. The optimization problem may then be formulated using the cost function

$$I = \frac{1}{2} \int_S (p(s) - p_{target}(s))^2 ds, \quad (21)$$

where S corresponds to the surface of the geometry and $p(s)$ refers to the pressure distribution over the surface.

For this case, an O-grid quadrilateral mesh has been used to numerically compute the flow over each airfoil. The mesh was uniformly refined as well as a function of the mesh level (ml). For instance, a mesh level of 1 corresponds to an O-grid of dimension 20×32 (32 elements around the airfoil and 20 elements in the normal direction to the far field), a mesh level of 2 corresponds to a grid of dimension 40×64 , and so on.

We consider now the inverse design of a NACA airfoil. In particular, in this case, a NACA 4412 airfoil is considered as the target profile, while a symmetric airfoil is treated as the initial profile to be optimized. A flow with a Mach number of $M = 0.1$ and an angle of attack of $\alpha = 0^\circ$ has been used. Moreover, a total of 12 design variables (the NURBS control points of the airfoil) have been used in the optimization to modify the airfoil shape. To demonstrate optimization at high orders for the NURBS-enhanced DG method, several different orders P have been considered. The drop in the cost function and its gradient, as a function of design iterations, as well as the initial and optimized airfoil profiles is shown in Figures (5) and (6).

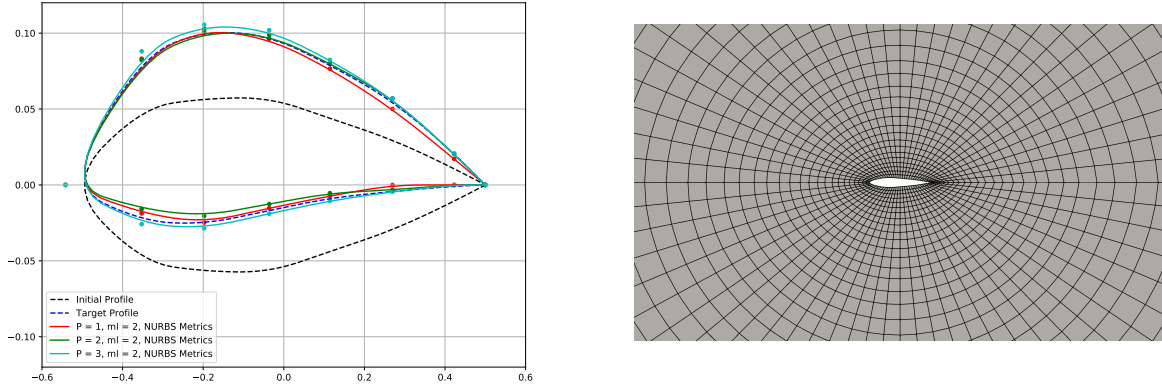


Fig. 5 The initial and optimized profiled (for various orders P) shown along with the control points used to parametrize the airfoils (left). The structured grid around the initial airfoil (right).

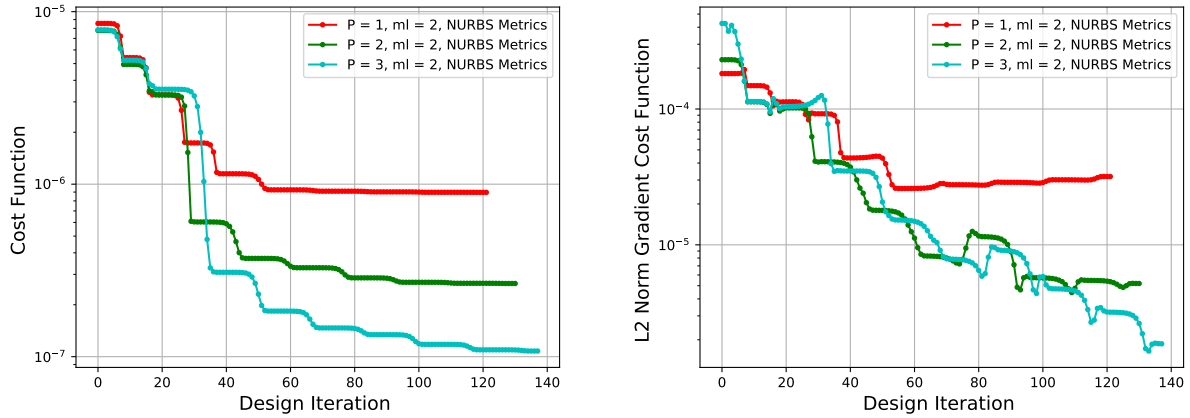


Fig. 6 Convergence of the cost function (left) and its gradient (right) for the inverse design case.

B. Airfoil Target Lift Coefficient - Unstructured Grid

The second case considered is a constrained target lift coefficient optimization case. In this optimization problem, a symmetric airfoil is used as the initial profile with a C_L of 0. The grid used for this case is moreover unstructured. We have generated it by meshing the parametric domain with triangles and then mapping each triangle onto the physical domain using the NURBS mapping. In this case, we aim to highlight the capability of the NURBS-enhanced approach to handle arbitrary element types.

For this optimization problem, a fixed target lift coefficient value of $C_{L,Target} = 0.25$ is specified, and the goal in the optimization is to be able to attain this specified lift. This is done through the minimization of the following functional:

$$I = \frac{1}{2}(C_L - C_{L,Target})^2. \quad (22)$$

For the cases considered, the optimization problem has been constrained using, first, an equality constraint on the pitching moment about the leading edge with a value of $C_m = -0.1255$ and, second, an equality constraint imposing no volume change in the airfoil throughout the optimization. As before, a flow with a Mach number of $M = 0.1$ and an angle of attack of $\alpha = 0^\circ$ has been used and a total of 12 design variables (the NURBS control points of the airfoil) have been used in the optimization to modify the airfoil profile. Figure (7) depicts the optimized airfoil profiles as well as the

unstructured grid around the initial symmetric airfoil. Moreover, Figure (8) depicts the optimization convergence results as a function of design iterations for various different orders P .

The applicability of the NURBS-enhanced approach to unstructured grids demonstrates its versatility to arbitrary element types while retaining its key property of maintaining an exact parametrization of the geometry. This highlights one of the key advantages of the NURBS-enhanced method relative to Isogeometric analysis (IGA). Introduced by Hughes, Cottrell and Bazilevs [14, 23], IGA consist of using NURBS basis functions to form the function space to represent the flow’s numerical solution. This, in turn, results in no approximation to the representation of the geometry in the analysis (the geometry can be represented exactly when solving the flow). In its basic setting, however, IGA is restricted to tensor-product splines and, hence, also meshes with solely tensor-product elements. As a result of this, there has been increased research in the past few years into unstructured spline spaces to improve the versatility of IGA. The NURBS-enhanced approach, on the other hand, is able to accommodate various element types with ease, as its implementation is independent of the element types present in the mesh.

VI. Conclusion

In this work, we have demonstrated adjoint based aerodynamic shape optimization for the NURBS-enhanced DG method. A key advantage of the NURBS-enhanced approach is in its exact parameterization of the geometry. The NURBS mapping is effectively used to compute all metric terms (which are then used to compute the Jacobian and normal vectors) in each element volume and on each element face, thus resulting in an accurate representation of the geometry. We furthermore investigated the efficacy of the shape optimization framework through two test cases. First, successful results to an inverse design problem involving a NACA 4412 airfoil as the target shape were presented. Following this, we considered the shape optimization of an airfoil to a target lift coefficient with constraints placed on the pitching moment and volume change.

These preliminary results illustrate the effectiveness of the NURBS-enhanced approach and its ability to bridge the gap between CAD and analysis. The versatility of the NURBS-enhanced method to handle arbitrary element shapes, and its ease of implementation, makes it advantageous to IGA (which also allows for an exact geometry parameterization). In future work, we aim first to investigate the extension of this approach to complicated geometries that require a multi-patch NURBS representation. Moreover, we will also consider how our method scales to higher dimensions and to more complicated viscous flows.

VII. Acknowledgment

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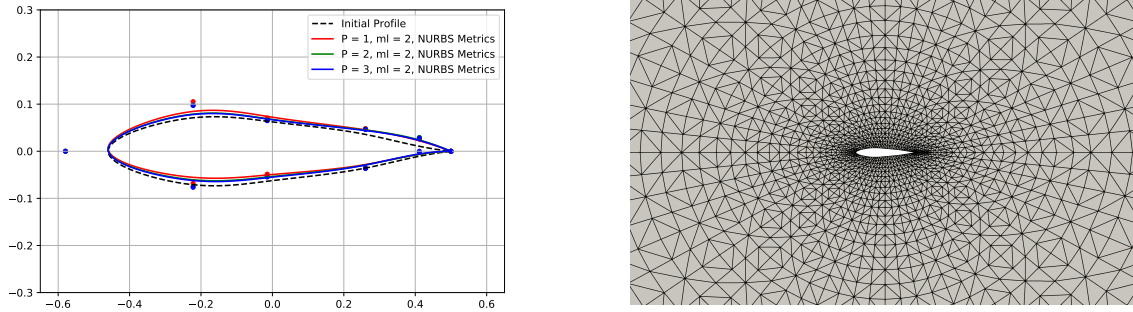


Fig. 7 The initial and optimized airfoil profiles for the target lift coefficient optimization case (left). The control points used for the optimization (to parametrize the airfoil) are also shown. Moreover, the unstructured grid around the initial airfoil is also shown (right).

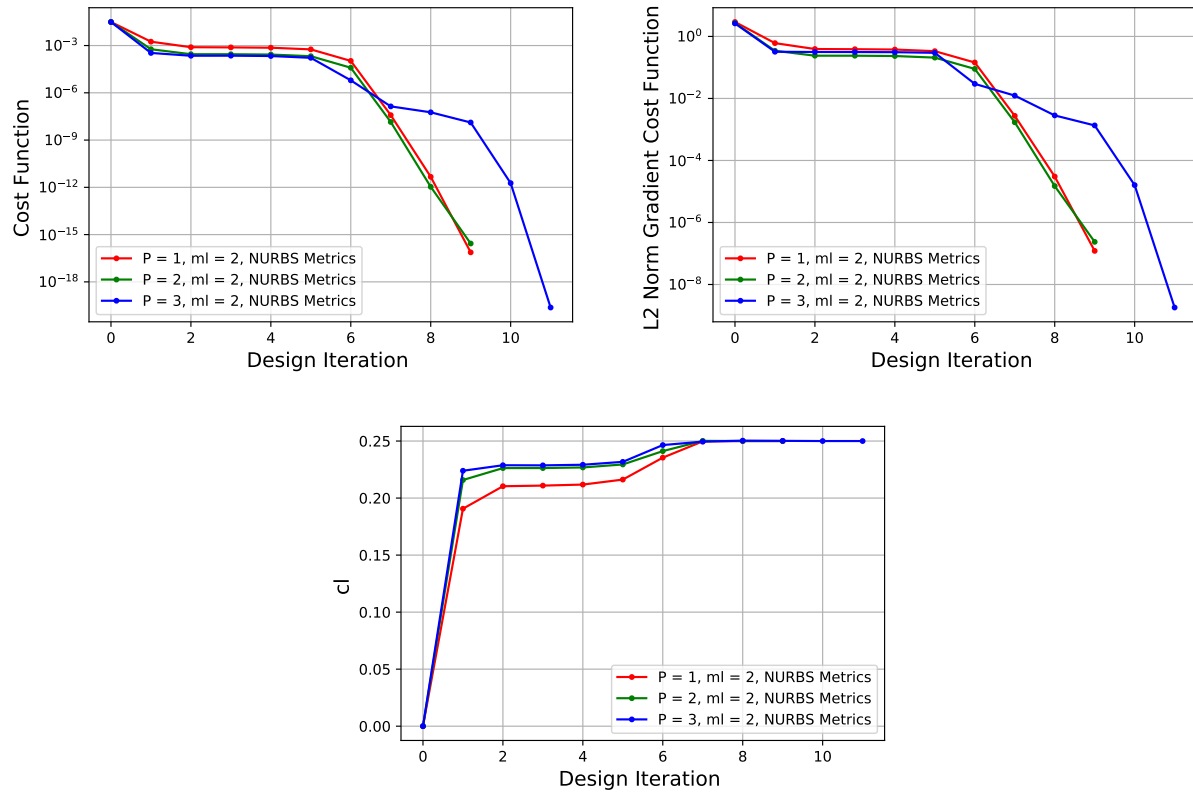


Fig. 8 Convergence of the cost function (top left), the L2 norm of the cost function's gradient (top right), and the lift coefficient (bottom) for the target lift coefficient optimization case on an unstructured grid.

References

- [1] Witherden, F. D., and Jameson, A., *Future Directions in Computational Fluid Dynamics*, American Institute of Aeronautics and Astronautics, 2017. doi:doi:10.2514/6.2017-3791, URL <https://doi.org/10.2514/6.2017-3791>.
- [2] Cockburn, B., and Shu, C.-W., "The local discontinuous Galerkin method for time-dependent convection-diffusion systems," *SIAM Journal on Numerical Analysis*, Vol. 35, No. 6, 1998, pp. 2440–2463.

- [3] Cockburn, B., and Shu, C.-W., “Runge–Kutta discontinuous Galerkin methods for convection-dominated problems,” *Journal of scientific computing*, Vol. 16, No. 3, 2001, pp. 173–261.
- [4] Kopriva, D. A., and Kolas, J. H., “A conservative staggered-grid Chebyshev multidomain method for compressible flows,” *Journal of computational physics*, Vol. 125, No. 1, 1996, pp. 244–261.
- [5] Liu, Y., Vinokur, M., and Wang, Z. J., “Spectral difference method for unstructured grids I: basic formulation,” *Journal of Computational Physics*, Vol. 216, No. 2, 2006, pp. 780–801.
- [6] Huynh, H. T., “A flux reconstruction approach to high-order schemes including discontinuous Galerkin methods,” *18th AIAA Computational Fluid Dynamics Conference*, 2007, p. 4079.
- [7] Wang, Z. J., and Gao, H., “A unifying lifting collocation penalty formulation including the discontinuous Galerkin, spectral volume/difference methods for conservation laws on mixed grids,” *Journal of Computational Physics*, Vol. 228, No. 21, 2009, pp. 8161–8186.
- [8] Vincent, P. E., Castonguay, P., and Jameson, A., “A new class of high-order energy stable flux reconstruction schemes,” *Journal of Scientific Computing*, Vol. 47, No. 1, 2011, pp. 50–72.
- [9] Wang, K., Yu, S., Wang, Z., Feng, R., and Liu, T., “Adjoint-based airfoil optimization with adaptive isogeometric discontinuous Galerkin method,” *Computer Methods in Applied Mechanics and Engineering*, Vol. 344, 2019, pp. 602–625.
- [10] Li, D., and Hartmann, R., “Adjoint-Based Error Estimation and Mesh Refinement in an Adjoint-Based Airfoil Shape Optimization of a Transonic Benchmark Problem,” *New Results in Numerical and Experimental Fluid Mechanics X*, Springer, 2016, pp. 537–546.
- [11] Chen, G., and Fidkowski, K. J., “Discretization error control for constrained aerodynamic shape optimization,” *Journal of Computational Physics*, 2019.
- [12] Nadarajah, S. K., *The discrete adjoint approach to aerodynamic shape optimization*, stanford university Stanford, USA, 2003.
- [13] Piegl, L., and Tiller, W., *The NURBS book*, Springer Science & Business Media, 2012.
- [14] Cottrell, J. A., Hughes, T. J., and Bazilevs, Y., *Isogeometric analysis: toward integration of CAD and FEA*, John Wiley & Sons, 2009.
- [15] Legrain, G., “A NURBS enhanced extended finite element approach for unfitted CAD analysis,” *Computational Mechanics*, Vol. 52, No. 4, 2013, pp. 913–929.
- [16] Sevilla, R., Fernández-Méndez, S., and Huerta, A., “3D NURBS-enhanced finite element method (NEFEM),” *International Journal for Numerical Methods in Engineering*, Vol. 88, No. 2, 2011, pp. 103–125.
- [17] Hesthaven, J. S., and Warburton, T., *Nodal discontinuous Galerkin methods: algorithms, analysis, and applications*, Springer Science & Business Media, 2007.
- [18] Kopriva, D. A., “Metric identities and the discontinuous spectral element method on curvilinear meshes,” *Journal of Scientific Computing*, Vol. 26, No. 3, 2006, p. 301.
- [19] Zwanenburg, P., and Nadarajah, S., “Equivalence between the energy stable flux reconstruction and filtered discontinuous Galerkin schemes,” *Journal of Computational Physics*, Vol. 306, 2016, pp. 343–369.
- [20] Jameson, A., “Aerodynamic design via control theory,” *Journal of scientific computing*, Vol. 3, No. 3, 1988, pp. 233–260.
- [21] Batina, J. T., “Unsteady Euler airfoil solutions using unstructured dynamic meshes,” *AIAA journal*, Vol. 28, No. 8, 1990, pp. 1381–1388.
- [22] Poirier, V., and Nadarajah, S., “Efficient reduced-radial basis function-based mesh deformation within an adjoint-based aerodynamic optimization framework,” *Journal of Aircraft*, Vol. 53, No. 6, 2016, pp. 1905–1921.
- [23] Hughes, T. J., Cottrell, J. A., and Bazilevs, Y., “Isogeometric analysis: CAD, finite elements, NURBS, exact geometry and mesh refinement,” *Computer methods in applied mechanics and engineering*, Vol. 194, No. 39-41, 2005, pp. 4135–4195.